Republic of the Philippines
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LAB REPORT

(DOUBLY LINKED LIST)

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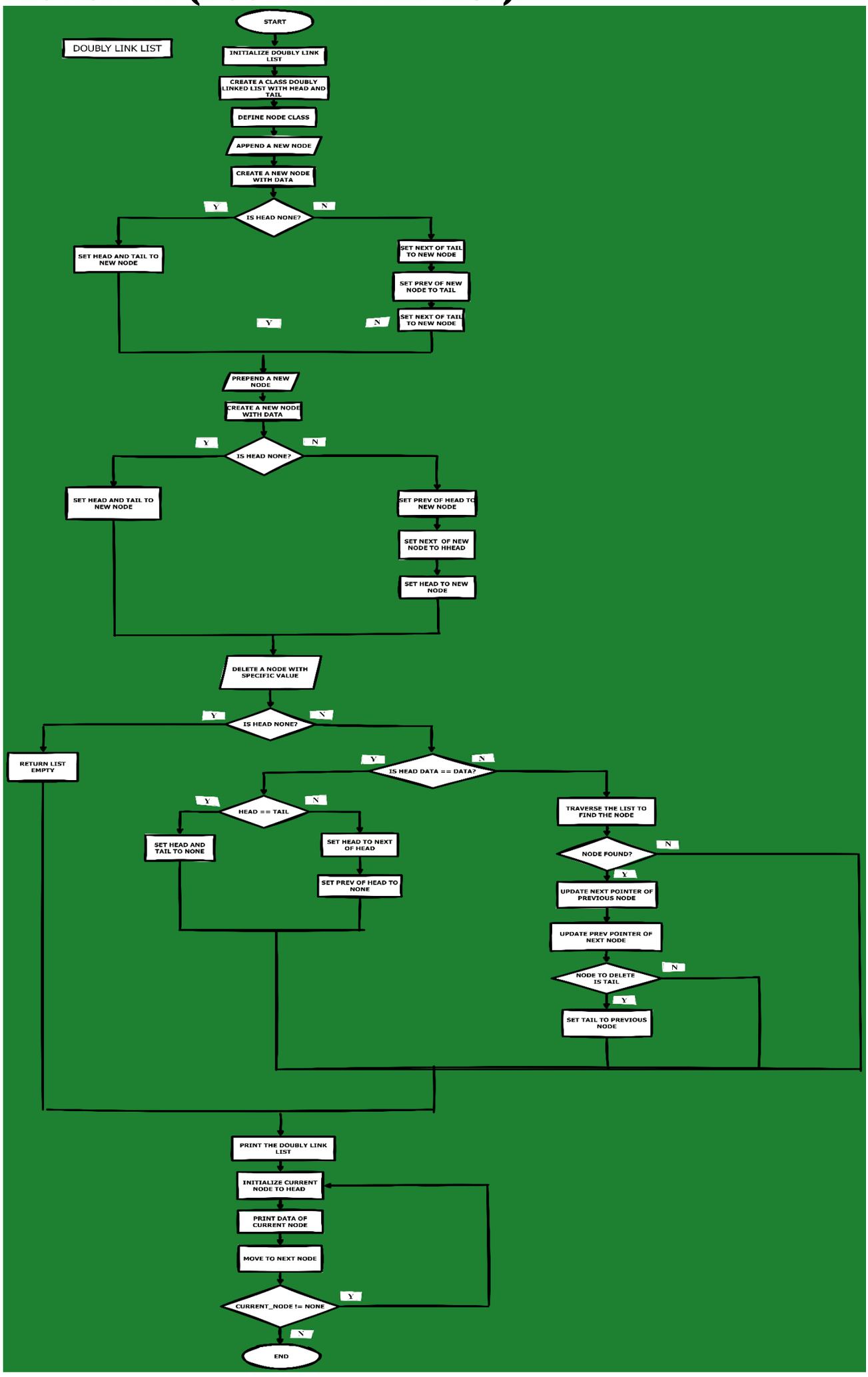
DATE:
DECEMBER 09,2024

DOUBLY LINKED LIST

MACHINE PROBLEM DESCRIPTION:

This program focuses on the operations of a Doubly Linked List, a data structure where each node contains data, a reference to the next node, and a reference to the previous node. This bidirectional linking allows for traversal in both forward and backward directions. The list starts with a head node and ends with a node pointing to None in both directions. Operations include adding nodes to the end (append), beginning (prepend), deleting nodes, and traversing the list to perform actions like printing data. This program helps us understand how Doubly Linked Lists work and their applications in solving real-world problems, particularly where bidirectional traversal is beneficial.

FLOWCHART (DOUBLY LINKED LIST):



SOURCE CODE:
The code below demonstrates the implementation of various operations on a Doubly Linked List of integers.

```python
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None
        self.prev = None

class DoublyLinkedList:
    def __init__(self):
        self.head = None
        self.tail = None

    def append(self, data):
        new_node = Node(data)
        if not self.head:
            self.head = new_node
            self.tail = new_node
            return
        self.tail.next = new_node
        new_node.prev = self.tail
        self.tail = new_node

    def prepend(self, data):
        new_node = Node(data)
        if not self.head:
            self.head = new_node
            self.tail = new_node
            return
        self.head.prev = new_node
        new_node.next = self.head
        self.head = new_node

    def delete_with_value(self, data):
        if not self.head:
            return
        if self.head.data == data:
            if self.head == self.tail:
                self.head = None
                self.tail = None
            else:
                self.head = self.head.next
                self.head.prev = None
            return
        current_node = self.head
        while current_node:
            if current_node.data == data:
                if current_node == self.tail:
                    self.tail = current_node.prev
                    self.tail.next = None
                else:
                    current_node.prev.next = current_node.next
                    current_node.next.prev = current_node.prev
                return
            current_node = current_node.next

    def print_list(self):
        current_node = self.head
        while current_node:
            print(current_node.data, end=" <-> ")
            current_node = current_node.next
        print("None")

# Example usage
dll = DoublyLinkedList()
dll.append(1)
dll.append(2)
dll.append(3)
dll.prepend(0)
dll.print_list()#TSHOW OUTPUT
dll.delete_with_value(2)
dll.print_list()  #OUPUT
```

OUTPUT:

The output displays the results of various operations on the Singly Linked List

```
[Running] python -u "c:\Users\Gian Carlo Trilles\OneDrive\Desktop\pythonfiles\DoublyLinkList.py"
0 <-> 1 <-> 2 <-> 3 <-> None
0 <-> 1 <-> 3 <-> None

[Done] exited with code=0 in 0.104 seconds
```